MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF MATHEMATICS AND STATISTICS

SYLLABUS

1. PROGRAMME	:	Bachelor of Science
2. SUBJECT	:	Statistics
3. LEVEL OF STUDY	:	2
4. COURSE TITLE	:	Introduction to statistical Analysis
5. COURSE CODE	:	STAT 2301
6. DURATION	:	16 Weeks
7. PRESENTED TO	:	Senate
8. PRESENTED BY	:	Department of Mathematics and Statistics
9. LECTURES PER WEEK	:	3
10. TUTORIAL HOURS PER WEEK	:	1
11. PRACTICAL HOURS PER WEEK	:	1
12. STUDENT INDEPENDENT LEARNING HOURS	:	12
13. TOTAL COURSE CREDITS	:	12
14. ASSESSMENT METHODS	:	At least 2 continuous assessment tests, One end of semester examination
15. ASSESSMENT WEIGHTING	:	40% Continuous assessment 60% End of semester Examination
16. AIM(S) OF THE COURSE	:	To introduce univariate methods of Statistical analysis.
17. LEARNING OUTCOMES	:	A successful learner from this

programme will be able to:

- Calculate probabilities, means and variances of some special univariate discrete and continuous distribution.
- Present and interpret data.
- Use moment and Probability generating functions to calculate moments and identify probability distribution.
- Carry out tests for means, difference between two means and sample proportions
- Use excel to work out proportions estimates and carryout statistical tests.

18. Topic of the Course :

- **Descriptive statistics**
  - Presentation of data
  - Measures of central tendency and dispersion

- **Elementary probability**
  - Probability of simple and compound events
- **Random Variables**
  - Discrete and continuous random variables
- **Discrete probability distributions**
  - Uniform, Bernoulli, Binomial, geometric, negative binomial and Poisson
- **Continuous distributions**
  - Uniform, exponential, normal
- **Sampling distribution**
  - Distribution of sample mean, proportion and variance
- **Point estimation**
  - Pooled estimators
  - Interval estimation
- **hypothesis testing**
  - Tests for means,
  - difference between means
  - sample proportions
  - sample variance
- **Chi-square tests**: goodness-of-fit and contingency table

- **ANOVA**
  - One-way and 2-way.
- **use of statistical software: Excel and SPSS.**

19. Prescribed Texts:

1. Mendenhall, W., Beaver R. G. and Beaver, B. M. (2009). Introduction to Probability and Statistics, 13th Ed. Brooke/Cole

20. Recommended Texts

1. Marques de Sá, Joaquim P. (2007). Applied Statistics Using SPSS, STATISTICA, MATLAB and R, Springer-Verlag.
2. Crawshaw, J and Chambers, J (1984), A Concise Course in A – Level Statistics, Stanley Thormes (Publisher Ltd.)

This course outline was presented to Senate on…………………………………………
and Approved on ……………………………………….